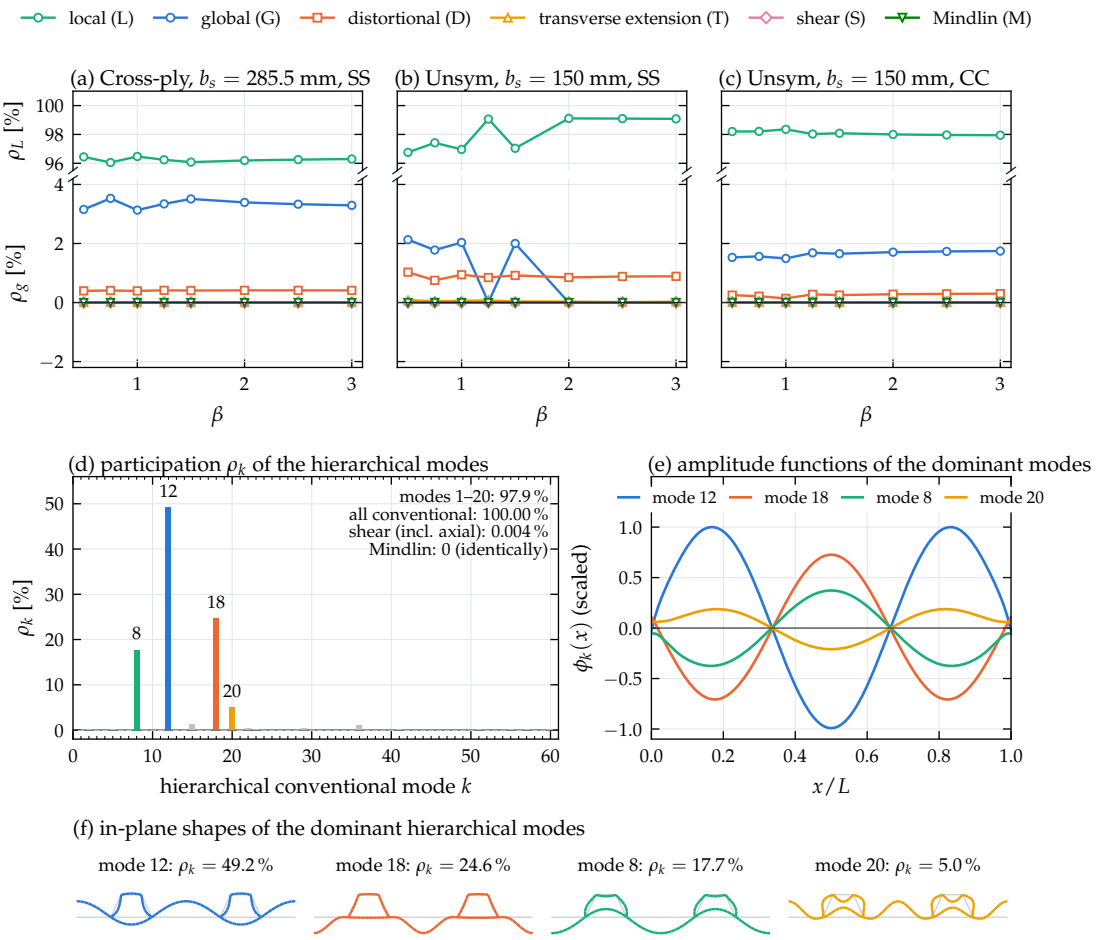

local (L)
global (G)
distortional (D)
transverse extension (T)
shear (S)
Mindlin (M)
(a) Cross-ply, $b_s = 285.5$ mm, SS
(b) Unsym, $b_s = 150$ mm, SS
(c) Unsym, $b_s = 150$ mm, CC
ρ_L [%]
ρ_g [%]
β
(d) participation ρ_k of the hierarchical modes
modes 1–20: 97.9 %
all conventional: 100.00 %
shear (incl. axial): 0.004 %
Mindlin: 0 (identically)
ρ_k [%]
hierarchical conventional mode k
(e) amplitude functions of the dominant modes
mode 12
mode 18
mode 8
mode 20
$\phi_k(x)$ (scaled)
x/L
(f) in-plane shapes of the dominant hierarchical modes
mode 12: $\rho_k = 49.2\,\%$
mode 18: $\rho_k = 24.6\,\%$
mode 8: $\rho_k = 17.7\,\%$
mode 20: $\rho_k = 5.0\,\%$